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24MA313

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Department of Mathematics

III Semester B.E. Degree: TEST-2

ENGINEERING MATHEMATICS-III

Duration: 1 Hour 15 Minutes | Date: 7-01-2026 | Max. Marks:30

NOTE: PART-A is compulsory.
PART-B have internal choice.

Branch: CSE & ISE

PART-A

Q.NO	CO	CD	PI	QUESTION	MARKS
1.	CO 4	1,2	1.2.1	Find the number of permutations of *a*, *b*, *c*, $\cdots$ *x*, *y*, *z* in which none of the patterns **SPIN**, **GAME**, **PATH** or **NET** occurs.	07
2.	CO 5	1,2	1.2.1	An apple, a banana, a mango and an orange are to be distributed to four boys B1, B2, B3, B4. The boys B1 and B2 do not wish to have apple, the boy B3 does not want banana or mango and B4 refuses orange. In how many ways the distribution can be made so that no boy is displeased?	08

PART-B

Q.NO	CO	CD	PI	QUESTION	MARKS
3.	CO 4	1,2	1.2.1	Find the number of permutations of the letters of the word **INSTITUTION** (i)How many of these are begin with I? *(ii)*How many of these are begin with I and end with *N (iii)* In how many arrangements three *T*'s are together (iv)In how many arrangements three *T*'s are not together	08
4.	CO 5	1,3	1.2.1	In how many ways can the integers 1, 2,3,……..10 be arranged in a line so that no even integer is in its natural place.	07
				OR	
5.	CO 4	1,3	1.2.1	How many distinct four-digit integers can one make from the digits 1, 3, 3, 7, 7 and 8?	08

6	CO 5	1,2	1.2.1	A machine that inserts letters into envelopes goes haywire and insert letters randomly into envelopes. What is the probability that in a group of 100 letters (i)no letter is put into the correct envelope? (ii)exactly 1 letter is put into the correct envelope? (iii)exactly 98 letters are put into the correct envelope? (iv)exactly 99 letters are put into the correct envelope? (v) all letters are put into the correct envelopes?	07

Course Outcome: At the end of the course the students will have the ability to	
CO 4	Apply the techniques of counting to identify the number of ways in which a given task can be accomplished without list all the possibilities explicitly
CO 5	Derive the generating function for the given situation and evaluate the required coefficient.

PI's	
1.2.1	Apply laws of natural science to an engineering problem

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24CS330

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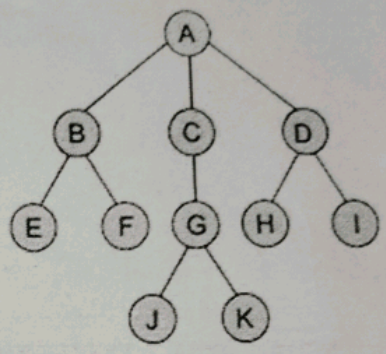
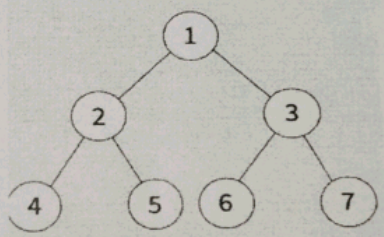
Department of Computer Science and Engineering

III Semester A ,C and E Sections: Test 2

Data Structures

Duration: 1hour 15 mins | Date: **08-01-2026** | Max. Marks: 30 Marks

NOTE: Question 1 is compulsory. Answer any one between question 2 and question 3.

Q.NO	CO	CD	PI	QUESTION	MARKS
1.	CO4	L2	1.4.1	a) For the tree given in the below figure, determine the following i) the path from A to node I ii) ancestors of node K iii) level number of node C and G iv) Degree of node E and F v) Out-degree of node A vi) descendants of node A 	6
				b) Find the inorder, preorder, postorder and level order traversals for the given tree 	4
				Construct the binary tree for the following expression i) ((a + b) – (c * d)) % ((e ^f) / (g – h)) ii) a + b / c * d – e	5
2.	CO2	L3	2.1.2	a) Write a C program to create a circular doubly linked list and perform insertions at the beginning and end of the list.	5+5+5
	CO3			b) Convert the following infix expression to postfix and trace the stack contents. (A + B) / (C + D) – (D * E) ^ F-G	

	CO3			c) Develop functions in C language to represent Priority Queue of integers with array representation. i) Insert(PQ, item): To insert item into a queue in ascending order, where PQ is a pointer to the Queue. ii) Delete(PQ): To delete highest prioritized item(Ascending order priority) in an unordered Queue where PQ is pointer to Queue.	
				OR	
3.	CO2	L3	2.1.2	a) Develop a C function to reverse the contents of doubly linked list with a header.	5+5+5
	CO3			b) Define recursion. Develop a C program that implements the solution of the tower of Hanoi problem.	
	CO3			c) Develop a C functions to insert at front and delete at rear end in a double ended queue.	

CO#	Course Outcomes	Highest Level of Cognitive Domain
CO1	Using the concept of pointers to arrays, structures, and functions, perform different operations on linear list with array storage representation.	L3
CO2	Use the concept of linked list representation to implement various operations on Linear lists with different types of linked list.	L3
CO3	Solve the problems of conversion and evaluation of different types of expressions using stack data structure, comprehend the implementation of different types of Queues.	L3
CO4	Using tree data structure solve the problems of searching, sorting, expression conversion and expression evaluation.	L3
CO5	Applying graph traversal techniques solve the problems of testing connectivity and acyclicity in a graph.	L3

PI's	
1.4.1	Apply theory and principles of computer science and engineering to solve an engineering problem
2.1.2	Identify processes/modules/algorithms of a computer-based system and parameters to solve a problem

Blooms Taxonomy	
L1	Remembering
L2	Understanding
L3	Applying



--- End ---

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"24CS350"

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JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Computer Science and Engineering
III Semester: CIE-3/Test-2

Discrete Mathematical Structures

Duration:1 Hr 15 min. Date:09.01.2026 Max. Marks:30

NOTE : PART- A is Compulsory, PART- B has internal choice

PART-A

Q.NO	CO	CD	QUESTION	MARKS
1	2	L3	Let A = {1,2,3} B={w,x,y,z} C = {4,5,6} Where R1⊆ AxB , R2 ⊆ BxC and R3 ⊆ BxC defined as R1 = {(1, w) (3, w) (2, x) (1, y)} R2= {(w, 5) (x, 6) (y, 4) (y, 6)} R3 = {(w, 4) (w, 5) (y, 5)} verify which of the following is TRUE a) R1 ∘ (R2∪R3) = (R1 ∘ R2) ∪ (R1 ∘ R3) b) R1 ∘ (R2∩R3) = (R1 ∘ R2) ∩ (R1 ∘ R3) c) M (R1 ∘ R2) = M(R1). M(R2) d) M(R1).M(R2)!= M(R2). M(R1)	08
2	4	L2	Explain the following with suitable example. a) Complement graph b) Subgraph c) Euler graph d) Hamiltonian graph e) Trail and path	07

PART-B

3	2	L3	Let R be a relation defined on A= {1,2,3,4,5,6,7,8,9,10,11,12} by $(x, y) \epsilon R$ if and only if $x - y$ is multiple of 5. a) Verify whether R is an equivalence relation. b) If yes, Find the partition of A induced by R.	07
4	4	L2	Define Isomorphism in graphs. Verify whether the graphs in Figure-1 are Isomorphic. If yes, map their edges and vertices.	08
			OR	
5	2	L3	Draw the Hasse diagram for the divisibility relation on a set A = {2,3,5,6,7,11,12,35,385} and find i) maximal and minimal element ii) greatest and least element iii) LUB {7,11} and LUB {5,7} iv) GLB {2,12} and GLB {2,3}	07
6	4	L2	Define Bipartite and Complete Bipartite Graphs with suitable example. Explain Kuratowski's graphs.	08

P.T.O

CO#	Course Outcomes	Highest Level of Cognitive Domain
CO1	Use the knowledge needed to test the logic of the program.	L3
CO2	Demonstrate the concept of relations and be able to determine their properties.	L3
CO3	Explain and apply decode group codes based on the various methods.	L3
CO4	Identify different types of graphs such as Isomorphism, Bipartite, planar, Hamiltonian paths and cycles, Euler trails and circuits based on their properties.	L3
CO5	Explain different types of trees and apply tree traversal techniques like preorder and postorder.	L3

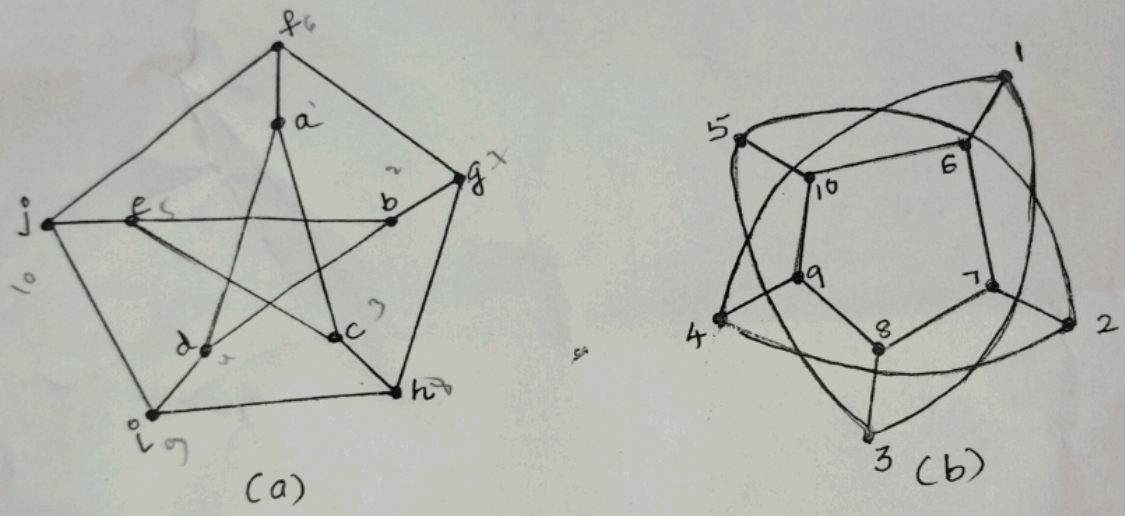


Figure - 1.

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24CS340

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JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Computer Science and Engineering

III Semester: CIE-2

COMPUTER ORGANIZATION

B, C & D Sections

Duration (In hrs): 1.15 Hrs | Date: 09/01/2026 | Max. Marks: 30

NOTE: Question 1 is compulsory, Question 2 and 3 has internal choice.

Q.No	CO	CD	PI	QUESTION	MARKS
1.	CO4	L3	1.4.1	a.Draw 4- bit carry look ahead adder and explain its operation with expressions. b.Perform the binary multiplication of (+14) and (-7) using Booth algorithm and 3-bit pair recoding.	9+6=15
2.	CO3	L2	2.3.1	a. Explain the structure of Static RAM cell with diagram. b. Briefly explain Direct Mapping Technique. c.Write the internal organization of 256M bit DRAM chip(32Mx8)	4+4+7 = 15
				OR	
3.	CO3	L2	2.3.1	a. Cache : 128 blocks : 16 words per blocks : 2048 words Main memory : 4k blocks : 16 words per block : 64k words i)Calculate the bits in MM address ii)Calculate the size of various fields in a)Direct b)2-way set c)4-way set d) Fully associative b. Explain the following terminologies i) Cache Hit ii) Cache miss iii) Replacement Algorithm	10+5 = 15

CO	Course Outcomes	Cognitive Domain
CO2	Explain the different ways of communicating with the I/O devices.	L2
CO3	Design and analyse the performance of memory systems.	L3

PI	
1.4.1	Apply theory and principles of computer science and engineering to solve an engineering problem
2.3.1	Able to apply computer engineering principles to formulate modules of a system with required applicability and performance

24CS320

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Computer Science and Engineering

III Semester BE: **Test-2**

Digital System Design

Duration: 1 Hr. 15 Min. **Date**: 08.1. 2026 **Max. Marks**: 30

NOTE: Question No. 1 is compulsory, answer either Question No. 2 or 3 completely.

Q.NO	CO	CD	PI	QUESTION	MARKS
1(a).	5	L2	2.3.1	Design mod 8 Synchronous Up Counter using JK Flip Flop with a neat diagram.	8
1(b).	4	L2	2.3.1	Explain SIPO and SISO shift registers with a neat diagram and truth table.	7
2(a).	2	L3	3.1.1	What are Multiplexers? Explain 4:1 MUX and 8:1 MUX with the necessary diagrams.	7
2(b).	3	L2	2.3.1	Explain Master Slave Flip flop using JK Flip flop with a neat timing diagram.	8
				OR	
3(a).	2	L3	3.1.1	Implement the following Boolean function using **8:1** MUX and **4:1** MUX for the following function: **$F(A,B,C,D)=\sum m(0,1,3,4,11,12,13,14,15)$**	7
3(b).	3	L2	2.3.1	Convert the following flip flops: i) SR flip flop to JK flip flop ii) SR to T flip flop to T flip flop	8

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Department of Computer Science and Engineering

III Semester: **Test II** **Section : C,D and E**

Object Oriented Programming using C++

Duration: 1.15 Hr Date: 07/01/2026 Max. Marks: 30 Marks

NOTE: Part A Question compulsory.

PART A

Q.No.	CO	CD	PI	QUESTION	MARKS
1 (a)	CO2	L3	2.1.2	Design a C++ class to represent a complex number using real and imaginary parts. Implement constructor overloading and operator overloading to add two complex numbers using the '+' operator. Demonstrate the working of the program by displaying the operands and the resultant complex number.	05
1(b)	CO2	L3	2.1.2	Design a class Counter with an integer member count. Implement the **prefix ++ operator** so that when it is applied to a Counter object, the count is incremented by 1. Write a program to show the operator in action.	05

1(c)	CO4	L3	2.1.2	Write a program which reads a text from the keyboard and displays the following information on the screen in **two columns**:
				(a) Number of lines
				(b) Number of words
				(c) Number of characters
				Note: Strings should be **left-justified** and numbers should be **right-justified** in a suitable field width.
				Output as follows:
				Output
				`Enter text (type 'END' on a new line to finish):`
				`This is a test`
				`Welcome to C++ programming`
				`Have fun coding`
				`END`
				
				`Description Count`
				`----------`
				`Number of lines 3`
				`Number of words 11`
				`Number of characters 55`
					05

2a	CO3	L3	2.1.2	Write a C++ program to define a class Student with data members name and rollNumber. Include a member function display() to print the student details. In the main() function, create an object of the Student class and use a **pointer to the object** to	07

				assign values to the data members and to call the display() function.	
2b	CO3	L3	2.1.2	Write a C++ program to demonstrate the use of the **this pointer**. Define a class Counter with an integer data member count. Implement a member function increment() that increments count and returns the current object using the this pointer. Also, include a function display() to show the value of count. In the main() function, create an object of the Counter class, call the increment() function multiple times, and display the final value.	08
			OR		
3a	CO3	L2	1.4.1	Explain multilevel inheritance in C++. Mention any two advantages of multilevel inheritance. Write a C++ program to demonstrate multilevel inheritance using three classes, where a derived class is created from another derived class.	07
3b	CO3	L2	1.4.1	Explain the following terms in C++: 1. Pointer to object 2. this pointer (Write brief explanations with examples where necessary.)	08

CO#	Course Outcomes	Highest Level of Cognitive Domain
CO1	Understand object-oriented programming concepts and be able to design solutions for a given problem.	L3
CO2	Able to reuse code with extensible Class types, user defined operators and function overloading.	L3
CO3	Achieve code reusability and extensibility by means of Inheritance and Polymorphism	L3
CO4	Incorporate file-handling functions and exceptions for a given application.	L3
CO5	Design C++ packages with templates for providing	L3

PI's	
1.4.1	Apply theory and principles of computer science and engineering to solve an engineering problem
2.1.2	Identify processes/modules/algorithms of a computer-based system and parameters to solve a problem

Blooms Taxonomy	
L1	Remembering
L2	Understanding
L3	Applying

--- End ---

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Department of Computer Science and Engineering

UNIVERSAL HUMAN VALUES

III Semester B.E: Test 2

Duration: 45 Mins. Date: 10.1.2026 Max. Marks: 15

NOTE: Answer any 2 full questions. Question 1 is compulsory

Q.NO	CO	CD	QUESTION	Marks
1	CO- 3	L2	Explain harmony in Human–Human relationships. Define Justice and discuss the nine universal values in relationships. Explain why Trust and Respect are considered the foundational values.	10
2	CO-4	L2	What is meant by harmony in Nature? Explain with examples.	5
3	CO-4	L2	Explain the four orders of Nature and their interconnectedness.	5

Course Outcomes (COs)

CO	Course Outcomes	Cognitive Domain
CO1	Understand self (I) clearly through self-exploration and demonstrate natural acceptance in situations by right understanding	L2
CO2	Distinguish between self (I) body, and co-existence of both to be in harmony.	L2
CO3	Relate to family and society by understanding nine universal human values in human- human relationship for harmony and prosperity.	L2
CO4	Explain interconnectedness and mutual fulfilment at all four orders of nature for co- existence.	L2
CO5	Develop commitment for holistic understanding of harmony by natural acceptance of human values in his/her profession and society.	L2

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create